

Trajectory Classification with LSTM for Robotic Peg-in-Hole

Group 1 Project

UE5 Apprentissage

Master's Machine Learning
Project – 2025–2026





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 - ▶ analyze the effect of modeling and data choices.
- ▶ Course link: the approach follows UE5 sequence modeling (RNN/LSTM module).



Roadmap (10 minutes)

1. Data extraction and preparation

Suggested pacing: ~ 3 min data+model, ~ 2 min plan+issues, ~ 3.5 min results, ~ 1.5 min wrap-up.

All figures are taken from project outputs in `images/`.



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- Files: `class1_trajectories.npy`, `class2_trajectories.npy`, `class3_trajectories.npy`

Observation block	Content	Size
peg_pose	position + quaternion	7
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Total features		20



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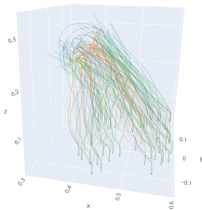
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- ▶ Numerical stability: replace zero std by small ϵ .



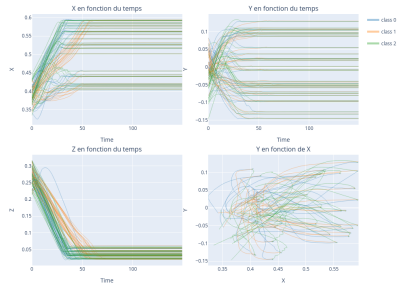
Trajectory Exploration Before Training

Trajectoires (position du peg) par classe



3D trajectory view (x, y, z) by class.

Trajectoires par composante



Temporal/component projections

$(x(t), y(t), z(t), y(x))$.

Initial separability patterns support using a temporal classifier rather than a static model.



Why LSTM for This Task?

- The class depends on **trajectory dynamics**, not one isolated timestep.

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- ▶ This is aligned with course motivation: preserve useful long-term information.



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- ▶ Optimizer: Adam(lr=1e-3)
- ▶ Batch size: 32, training budget: up to 40 epochs

Experimental Plan for Hyperparameters



Planned Ablations

Run	Main setting	Purpose
R1	hidden=100, z-score, 10 epochs	baseline behavior and convergence
R2	hidden=64, no normalization	test sensitivity to scaling choices
R3	hidden=64 + dropout	measure regularization effect
R4	R3 + gradient clipping	stabilize training dynamics

Additional fixed controls: seed, train/val/test split, optimizer family, and class definition.



Evaluation Protocol

- Primary metric: **test accuracy** on held-out internal test set (78 trajectories).

Interpretation caution: final numbers are taken from documented run logs (`rapport.ipynb/rapport_final.ipynb`); external teacher test remains the final robustness check.



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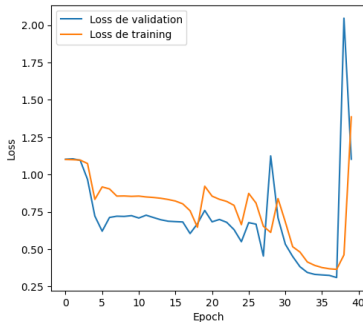
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- ▶ Supporting metrics: train/val loss and train/val accuracy across epochs.
- ▶ Selection logic: keep the simplest setting that maximizes generalization and stability.
- ▶ Assignment requirement considered: model interface kept compatible with teacher-provided unseen test tensor format.

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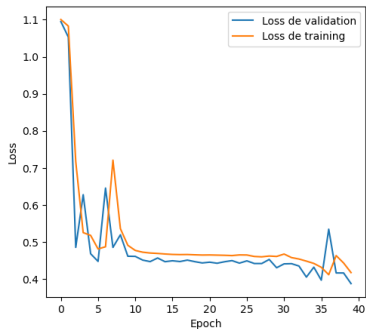


Practical Issues During Development

- **Reproducibility:** without fixed seeds, run-to-run comparison was unreliable.



Without normalization: noisier convergence.

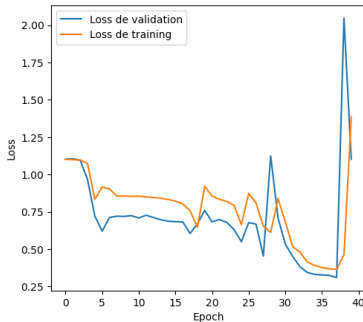


Without dropout: weaker regularization behavior.

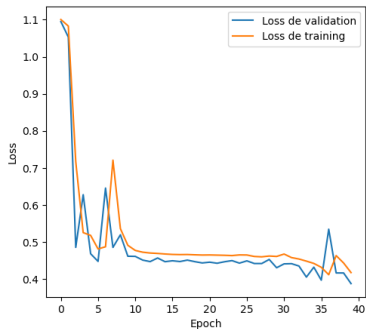


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- **Optimization sensitivity:** preprocessing and regularization strongly changed convergence behavior.



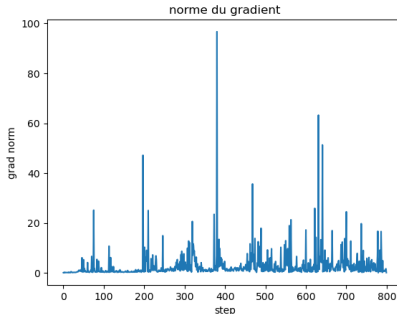
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Exploding Gradients and Fix



Observed spikes in gradient norm across training steps.

Problem

Large gradient spikes destabilized updates.

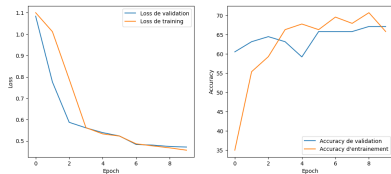
Mitigation

Gradient clipping in the training loop with max norm 1.0:

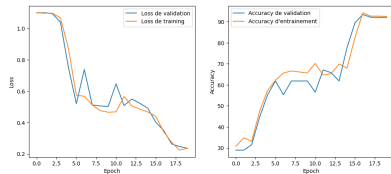
PyTorch `clip_grad_norm(...)`.



Runs R1 and R2



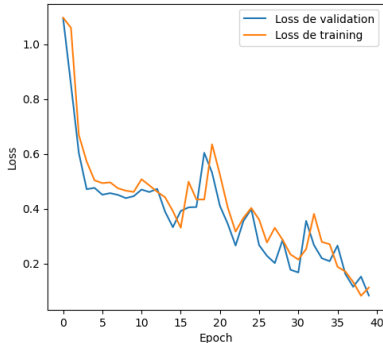
R1: baseline, limited performance and unstable validation behavior.



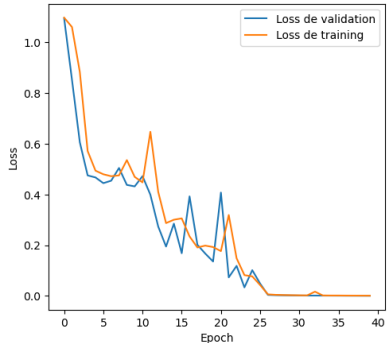
R2: major improvement, **92.31%** test accuracy (72/78).



Runs R3 and R4



R3 (dropout): run logs report **98.72%** test accuracy (77/78).



R4 (dropout + clipping): run logs report the best result, **100%** (78/78).



Quantitative Summary and Interpretation

Run	Test accuracy	Key effect
R1	low baseline	underfit/unstable baseline
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- ▶ Most impactful additions: **dropout** + **gradient clipping**.
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- ▶ Internal split results are excellent; external hidden test set is still required for final robustness validation.



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- ▶ In the documented experiment logs, the final recipe (2-layer LSTM, dropout, clipping) reached perfect internal test performance.
- ▶ The experiment plan made performance gains explainable, not only empirical.
- ▶ Main engineering lesson: stability controls are critical for recurrent models.



Next Steps

- Run official teacher-held test evaluation to confirm out-of-sample robustness.

Material sources: `classification.ipynb`, `rapport_final.ipynb`,
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- ▶ Complete the assignment analysis on **feature influence** (e.g., pose-only, velocity-only, reduced subsets).
- ▶ Report class-wise metrics and confusion matrix on external test for deeper error analysis.

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